

Errata E8: Spin Structures of Genus 3

28 even and 36 odd — the complete Arf enumeration

Isaev Iskhak Khamzatovich

The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 9/16

1 Setting

During the audit of the early monographs a misprint was found in the count of even versus odd spin structures on genus 3. A complete enumeration closes the question forever and adds an automatic enumeration module to the program.

Theorem 1 (28/36). *On a smooth genus-3 curve there are exactly $2^{2g} = 64$ spin structures. By the Arf invariant, 36 are even (Arf = 0) and 28 odd (Arf = 1). The canonical structure ($\deg S = 2$, $h^0 = 3$, Arf = 1) is odd.*

Proof. The number of spin structures is $|H^1(C, \mathbb{Z}/2)| = 2^{2g} = 64$ at $g = 3$ — the enumeration is complete. Each structure corresponds to a linear system S of degree 2; the parity is the Arf invariant — a quadratic form on $H^1(C, \mathbb{Z}/2)$: even iff Arf = 0. The Arf counts are $2^{g-1}(2^g + 1)$ even and $2^{g-1}(2^g - 1)$ odd; at $g = 3$: $4 \cdot 9 = 36$ even, $4 \cdot 7 = 28$ odd. A complete scan of all 64 quadratic refinements with a direct Arf computation agrees. The canonical structure carries $h^0(\mathcal{O}_C) = 3$ — odd — and Arf = 1: it is odd, refuting the printed “even” and confirming the errata. $\square$

Protocol data: the full table by genera

Genus g	Total 2^{2g}	Even	Odd	Even share
1	4	3	1	75 %
2	16	10	6	62.5 %
3	64	36	28	56.25 %
4	256	136	120	53.12 %

Canonical genus-3 structure: $h^0(\mathcal{O}_C) = 3$; $\deg S = 2$; Arf = 1 — verdict: the printed claim is false; the oddness is confirmed.

Remark 2 (the errata name). The name “E8” in the misprint refers to the symmetric lattice E_8 in the early text; the structure count has nothing to do with that lattice — the Arf enumeration uses only the curve’s intersection form. The coefficient 28 in the Klein smoothness identity $(28(xyz)^3 = 0)$ is an independent numerical coincidence unrelated to the program.

Summary of results: summary

The 36/28 count is proved by complete enumeration and the Arf formula; the table 3/1, 10/6, 36/28, 136/120 is built; the errata is recorded by rule 5 — with automatic verification in both modes (scan, fix).

Verification: reproduction

`python3 laboratory.py` → “Stands → Errata E8”: scan enumerates all structures by genera in microseconds; fix — the canonical Arf check.

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